

Particular Solution: Case-ii: $f(n)$ is in the form of β^n , where β is not a characteristic root, then the particular solution to be assumed as $A\beta^n$

Find the solution of the recurrence relation

$$a_n - 5a_{n-1} + 6a_{n-2} = 7^n$$

Solution: $a_n - 5a_{n-1} + 6a_{n-2} = 7^n$

the solution of above recurrence relation is

$$a_n = a_n^h + a_n^p$$

a_n^h can be find by assuming $a_n - 5a_{n-1} + 6a_{n-2} = 0$
the characteristic equation of given recurrence relation is

$$\alpha^2 - 5\alpha + 6 = 0 \Rightarrow \alpha = 2, 3$$

$$\therefore a_n^h = A_1(2)^n + A_2(3)^n$$

The a_n^p of $a_n - 5a_{n-1} + 6a_{n-2} = 7^n$ can be find
by assuming $a_n^p = A \cdot 7^n$

$$\Rightarrow a_{n-1}^p = A \cdot 7^{n-1}$$

$$\Rightarrow a_{n-2}^p = A \cdot 7^{n-2}$$

$$a_n - 5a_{n-1} + 6a_{n-2} = 7^n$$

$$\Rightarrow A \cdot 7^n - 5 \cdot A \cdot 7^{n-1} + 6 \cdot A \cdot 7^{n-2} = 7^n$$

$$\Rightarrow A \cdot 7^n - \frac{5}{7} A \cdot 7^n + \frac{6}{49} A \cdot 7^n = 7^n$$

$$\Rightarrow 49A \cdot 7^n - 35 A \cdot 7^n + 6 A \cdot 7^n = (49) 7^n$$

$$\Rightarrow 20A \cdot 7^n = (49) 7^n$$

$$\Rightarrow 20A \cdot 7^n = (49) 7^n$$
$$\Rightarrow a_n^p = \frac{49}{20} 7^n$$

the solution of above recurrence relation is

$$a_n = a_n^h + a_n^p$$

$$\therefore a_n = A_1 (2)^n + A_2 (3)^n + \frac{49}{20} 7^n$$

Particular Solution: Case-iii: $f(n)$ is in the form of $(a + bn + cn^2 + dn^3 + \dots + en^r)\beta^n$, where β is not a characteristic root, then the particular solution to be assumed as $(A + Bn + Cn^2 + Dn^3 + \dots + En^r)\beta^n$

Find the particular solution of the recurrence relation

$$a_n - a_{n-1} = (3)(n)2^n$$

Solution: $a_n - a_{n-1} = (3)(n)2^n$

the solution of above recurrence relation is

$$a_n = a_n^h + a_n^p$$

a_n^h can be find by assuming $a_n - a_{n-1} = 0$ the characteristic equation of given recurrence relation is

$$\alpha - 1 = 0 \Rightarrow \alpha = 1$$

$$\therefore a_n^h = A_1(1)^n = A_1$$

The a_n^p of $a_n - a_{n-1} = (3)(n)2^n$ can be found by assuming $a_n^p = (A + Bn)2^n$
 $\Rightarrow a_{n-1}^p = (A + B(n-1))2^{n-1}$

Therefore the solution of $a_n - a_{n-1} = (3)(n)2^n$ is
 $A2^n + Bn(2^n) - A2^{n-1} - Bn(2^{n-1}) + B2^{n-1} = (3)(n)2^n$
 $\left(A - \frac{A}{2} + \frac{B}{2}\right)2^n + \left(B - \frac{B}{2}\right)n2^n = (3)(n)2^n$

$$\Rightarrow B = 6$$

$$\Rightarrow A = -6$$

$$\therefore \Rightarrow a_n^p = (-6 + 6n)2^n$$

the solution of above recurrence relation is

$$a_n = a_n^h + a_n^p$$

$$\therefore a_n = A_1 + (-6 + 6n)2^n$$

Particular Solution: Case-iv: $f(n)$ is in the form of $(a + bn + cn^2 + dn^3 + \dots + en^r)\beta^n$, where β is a characteristic root of multiplicity $(m-1)$, then the particular solution to be assumed as $(n^{m-1})(A + Bn + Cn^2 + Dn^3 + \dots + En^r)\beta^n$

Find the particular solution of the recurrence relation

$$a_n - 3a_{n-1} = (2)3^n$$

Solution: $a_n - 3a_{n-1} = (2)3^n$

the solution of above recurrence relation is

$$a_n = a_n^h + a_n^p$$

a_n^h can be find by assuming $a_n - 3a_{n-1} = 0$ the characteristic equation of given recurrence relation is

$$\alpha - 3 = 0 \Rightarrow \alpha = 3$$

$$\therefore a_n^h = A_1(3)^n$$

The a_n^p of $a_n - 3a_{n-1} = (2)3^n$ can be found by assuming $a_n^p = (An)3^n$
 $\Rightarrow a_{n-1}^p = (A(n-1))3^{n-1}$

Therefore the solution of $a_n - 3a_{n-1} = (2)3^n$ is

$$(An)3^n - 3(A(n-1))3^{n-1} = (2)3^n$$
$$An3^n - An3^n + A3^n = (2)3^n$$
$$\Rightarrow A = 2$$

$$\therefore \Rightarrow a_n^p = (2n)3^n$$

Therefore the solution of $a_n - 3a_{n-1} = (2)3^n$ is
$$a_n = A_1(3)^n + (2n)3^n$$

Find the particular solution of the recurrence relation

$$a_n - 3a_{n-1} = 2n, a_1 = 3$$